

ENGG1910 Demystifying AI

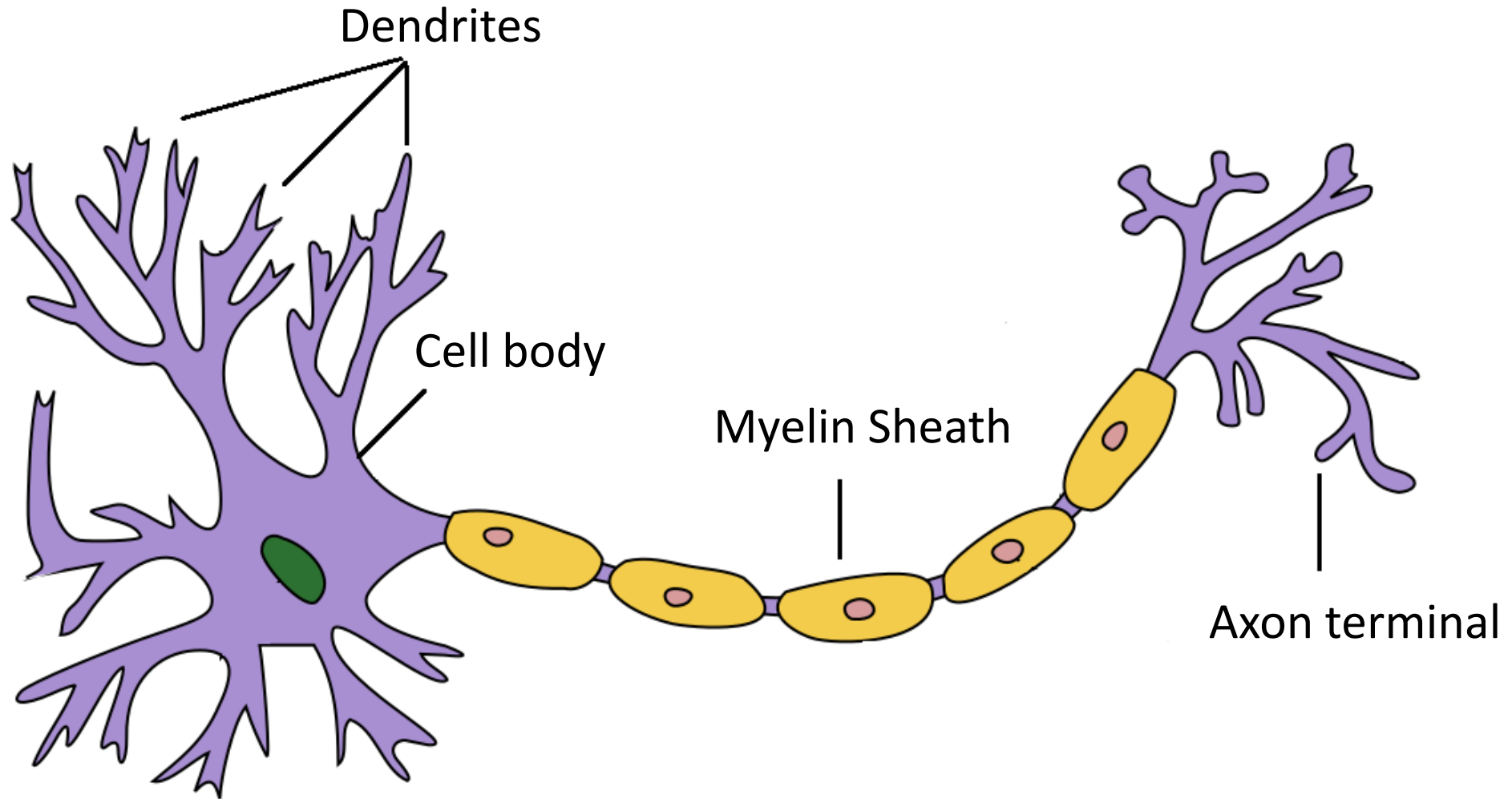
Lab2 Guidelines

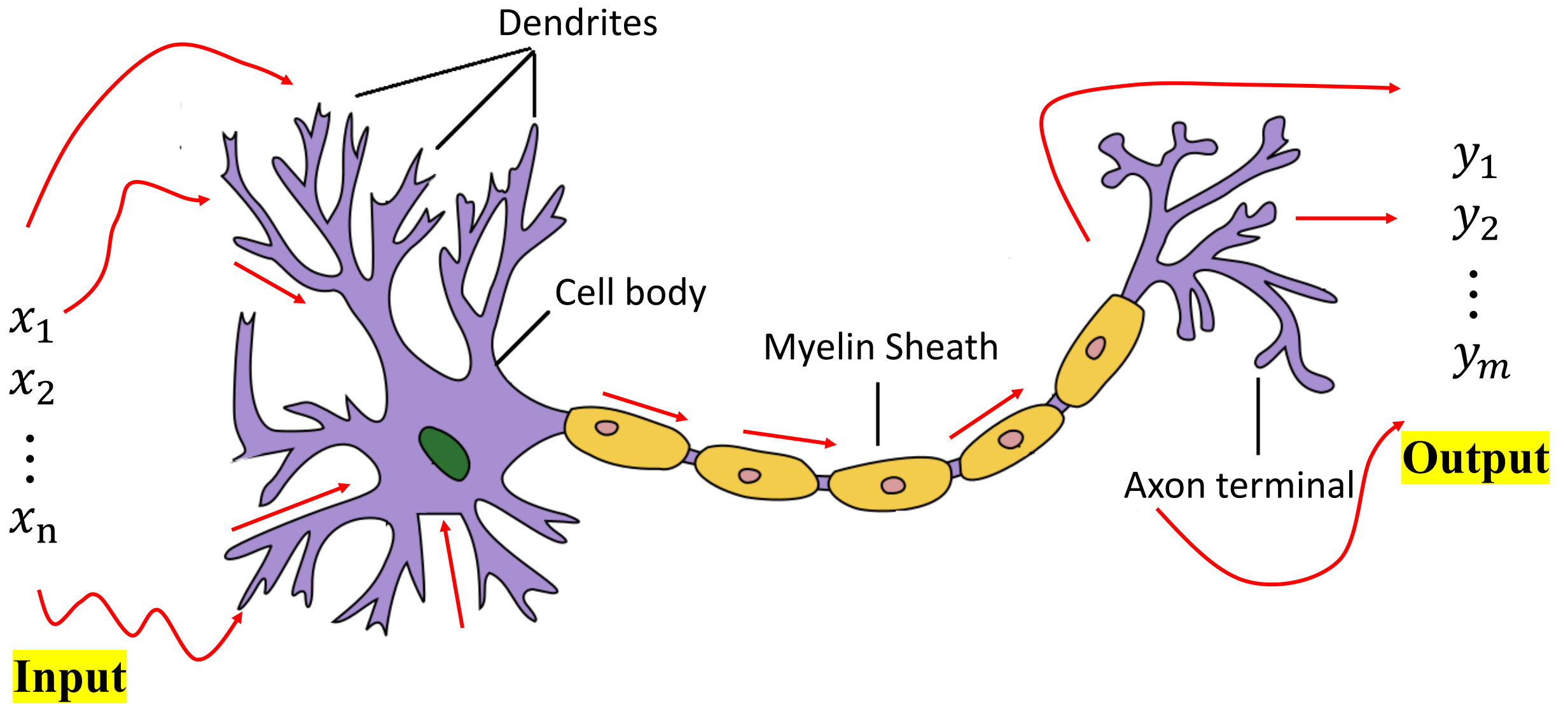
Objectives

- In this lab, students will learn the procedures to build a **Neural Machine Translation** (NMT) model in a toy example of *man2can*
- The concepts of **pre-trained model** and **fine-tuning** will be introduced. Students will learn to fine-tune a pretrained model with custom datasets
- Students are expected to understand which factors could affect the model performance

Network structure

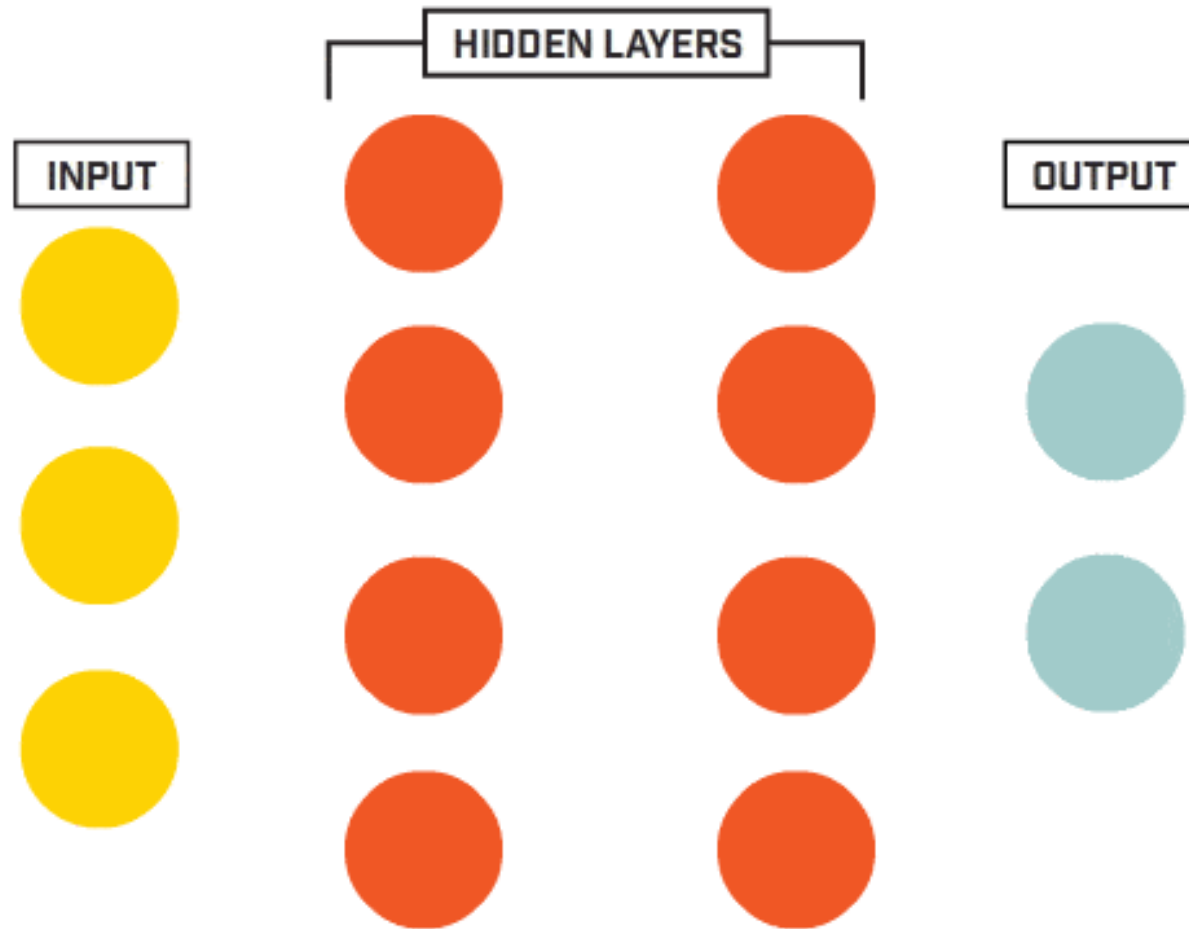
Here, we have a rough review of Biological neuron model.





Network structure

The different layers of the MLP neural network are fully connected.



A Neural Network

A system using **data**
to **answer questions**



An example: Wine or beer?



A neural network is a model

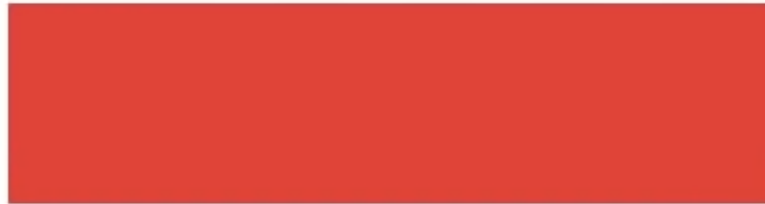


Model



Beer

Data collection



COLOR

13.5% Alc/volume

ALCOHOL

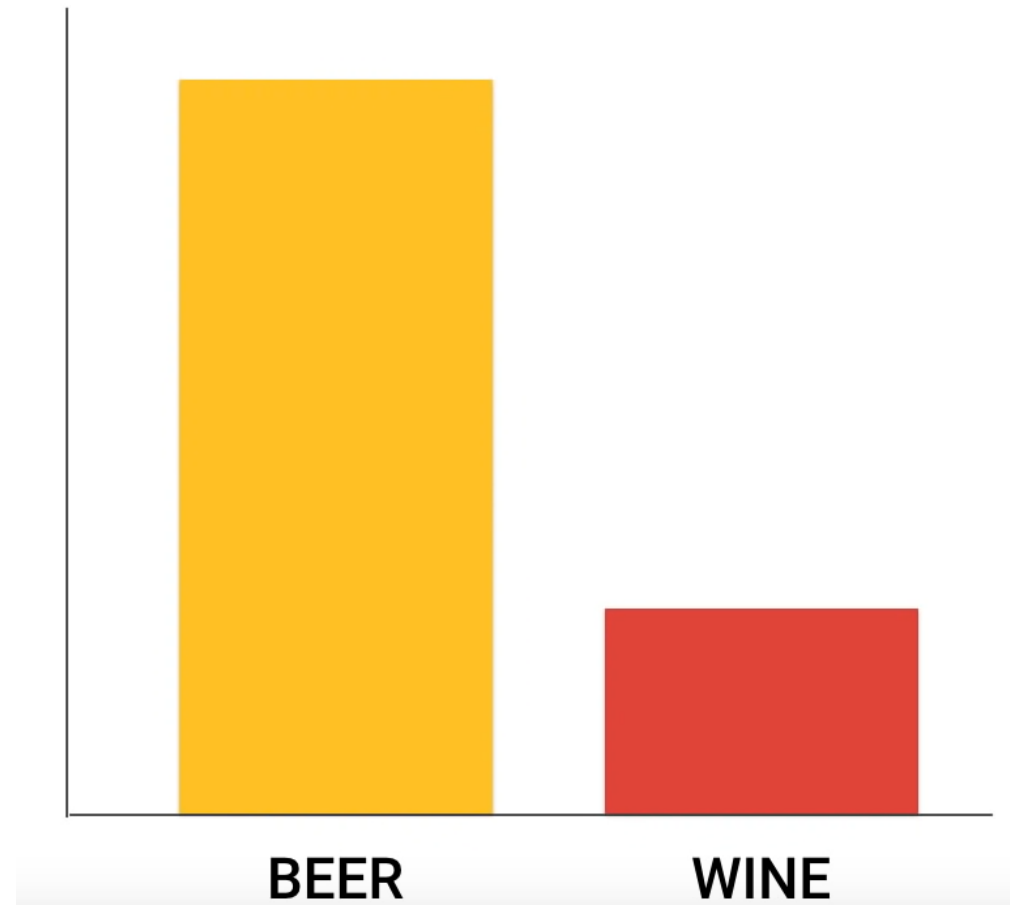
Data of key features



Data preparation

Color (nm)	Alcohol %	Beer or Wine?
610	5	Beer
693	14	Wine
426	17	Wine

Data preparation



Data preparation

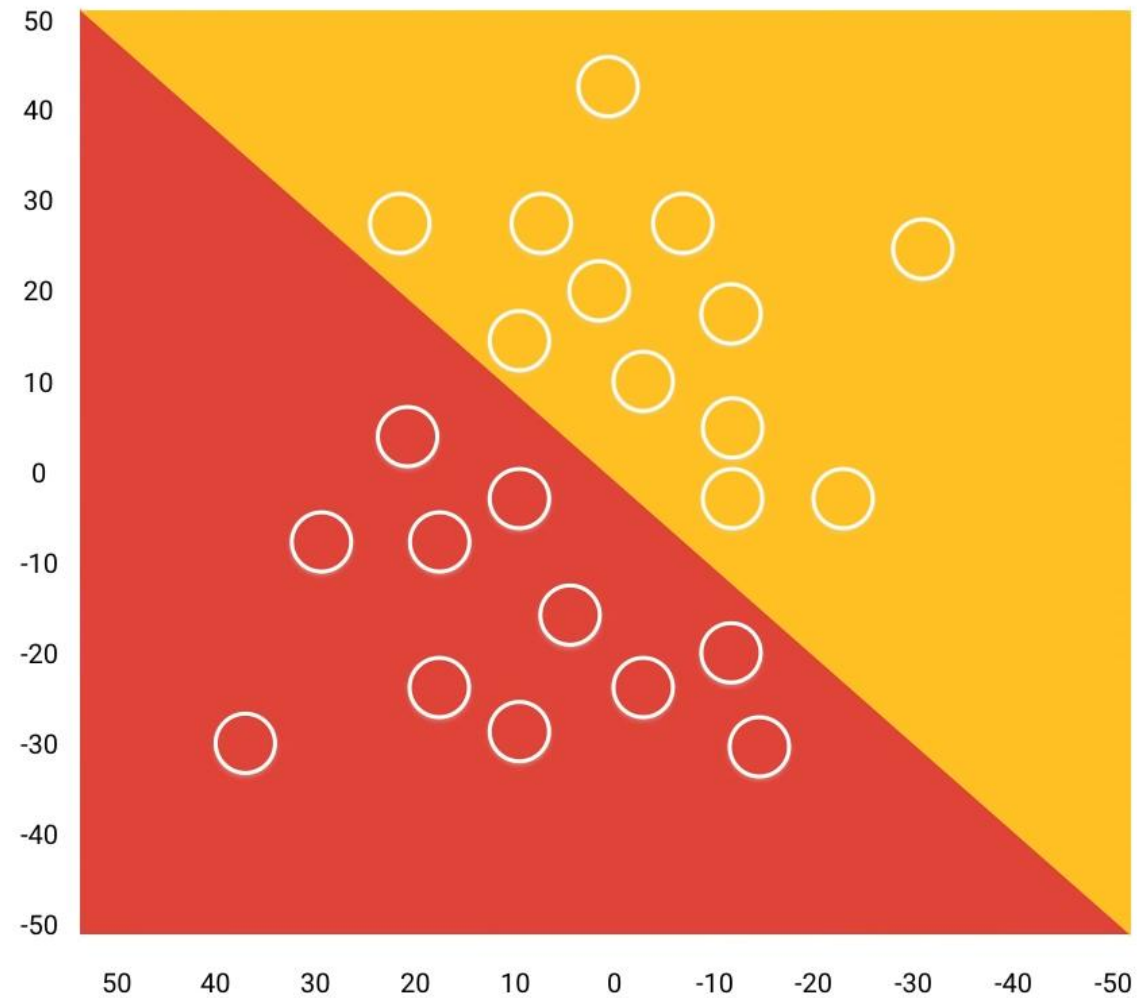


Training



Evaluation

Choosing a model



Training



Model

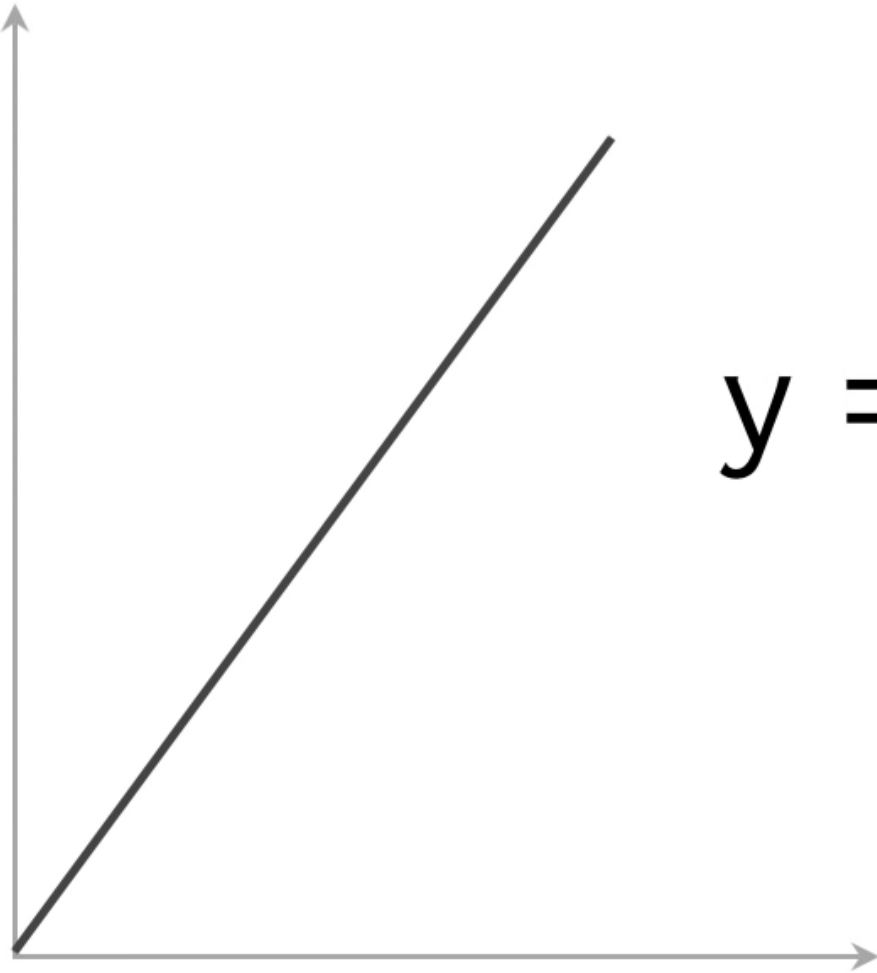


Beer

Training



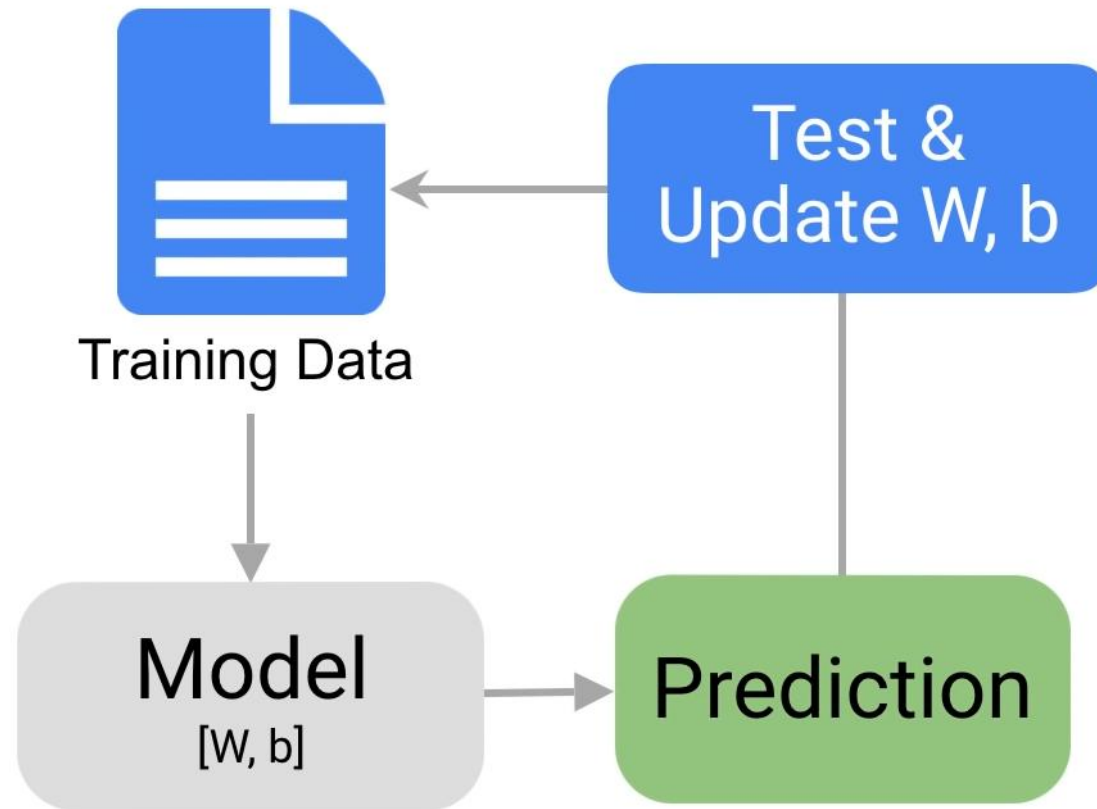
Training



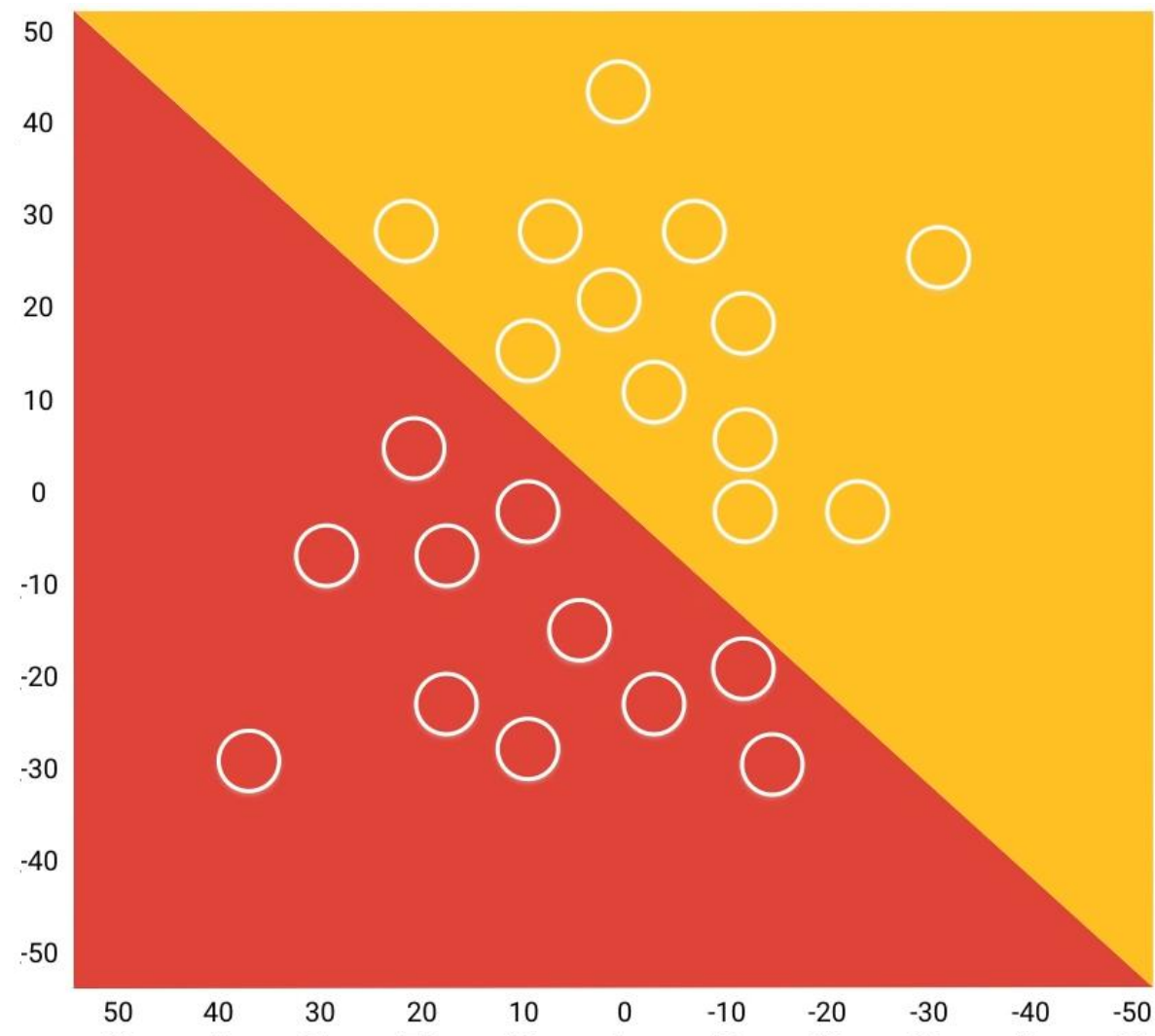
$$y = m * x + b$$

SLOPE INPUT Y-INTERCEPT

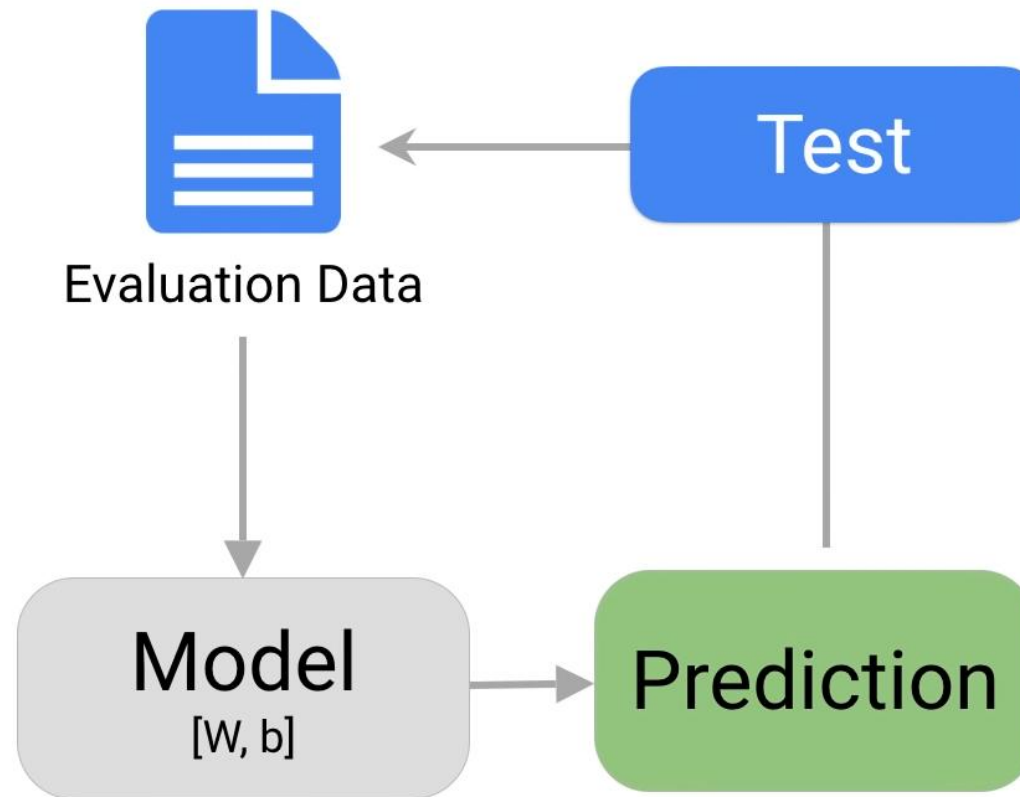
Training



Training



Evaluation



Evaluation

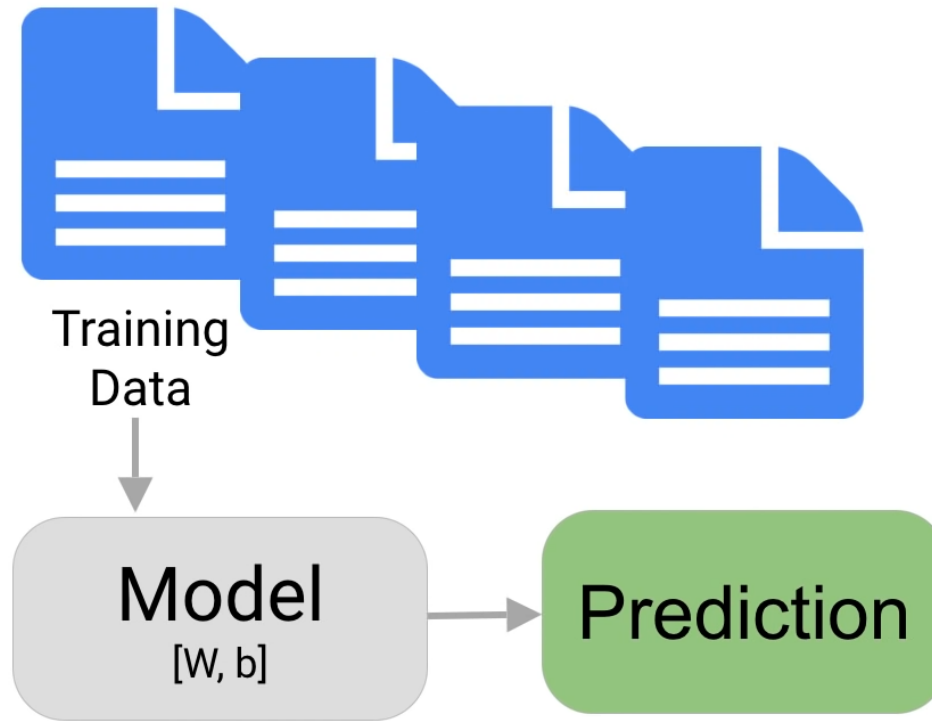


Training
80%



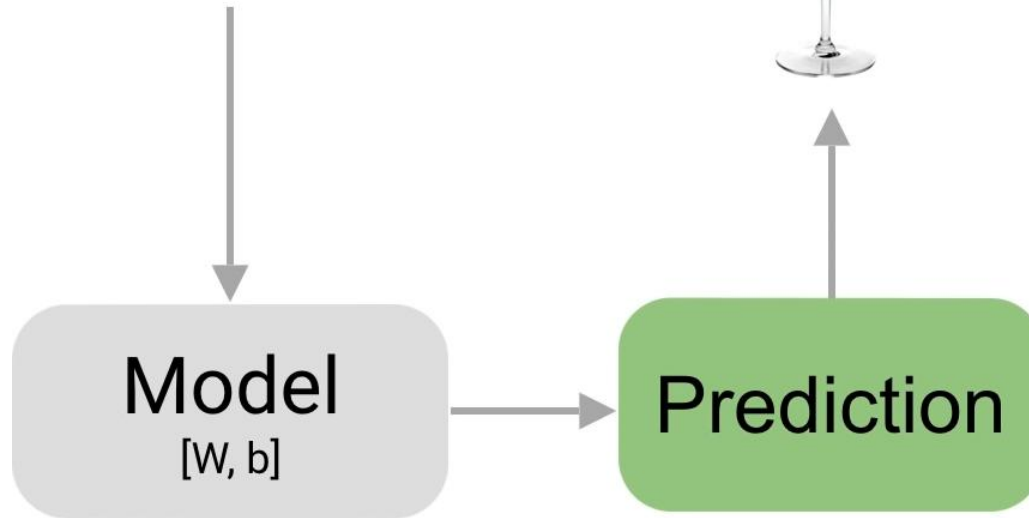
Evaluation
20%

Parameter tuning



Prediction

Color: 660nm
Alcohol: 12%



7 steps of AI learning

- Gathering data
- Preparing data
- Choosing a model
- Training data
- Evaluation
- Parameter tuning
- Prediction



Colab setup

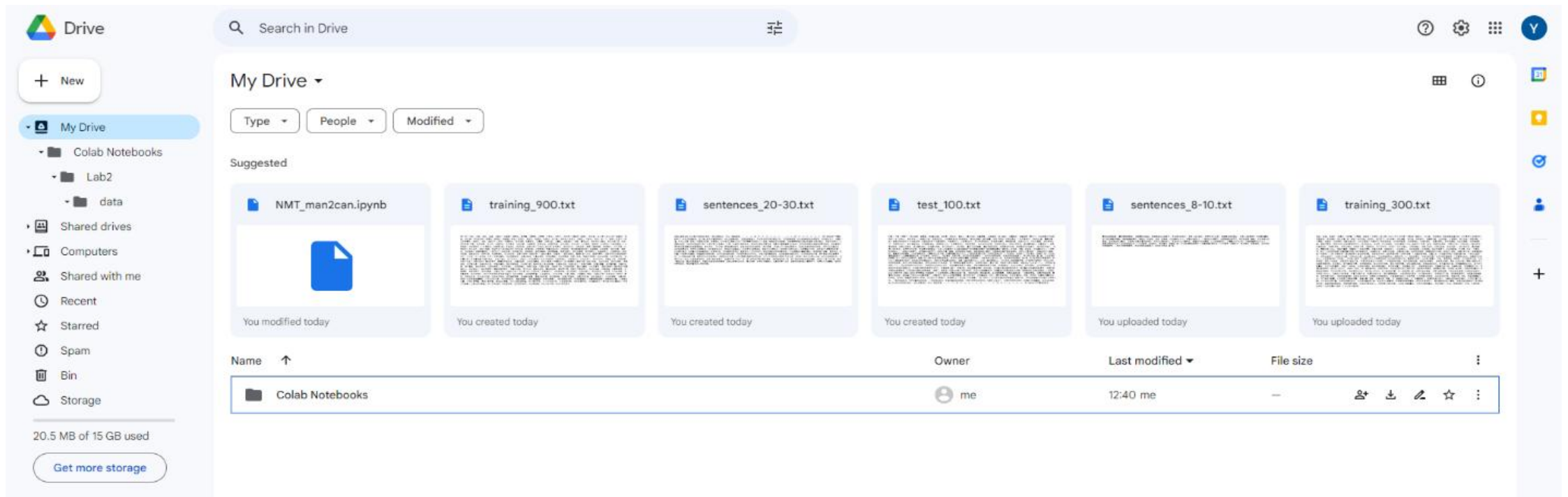
With Colaboratory (Colab), you can write and execute Python code in your browser, and:

- No configuration required
- Free use of GPU
- Easy sharing



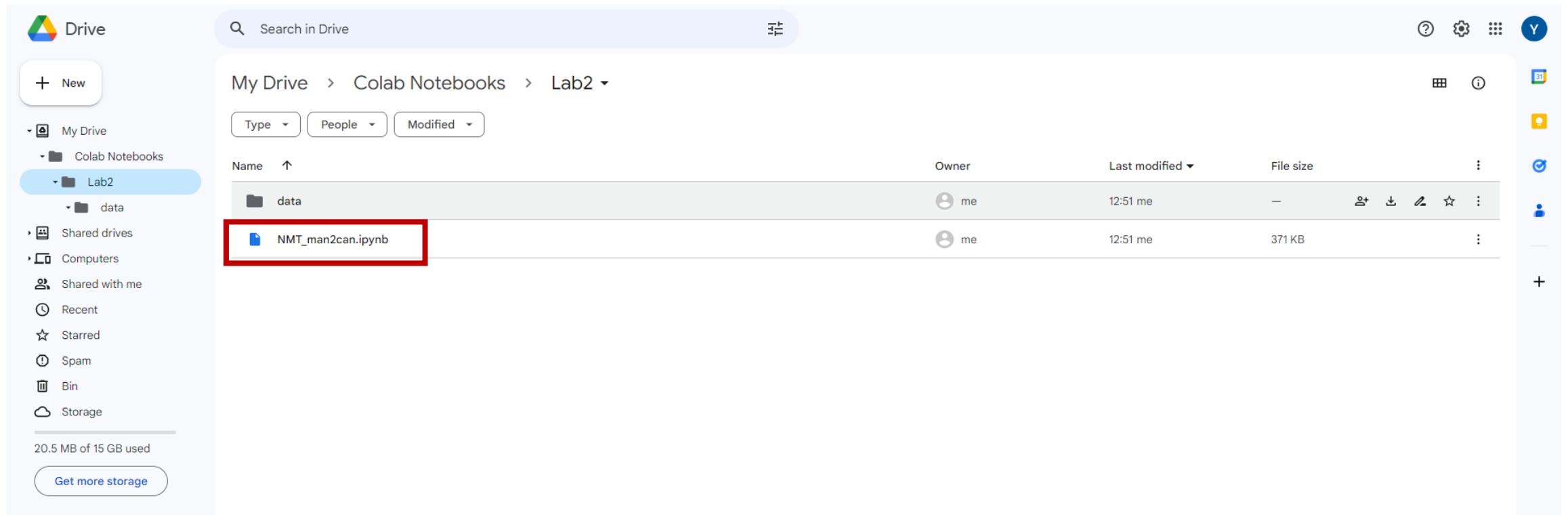
Prepare the workplace on Colab

- Log in to Google Drive using your Google account. Unzip the lab file and upload the “Colab Notebooks” folder to the root directory of your Google Drive.



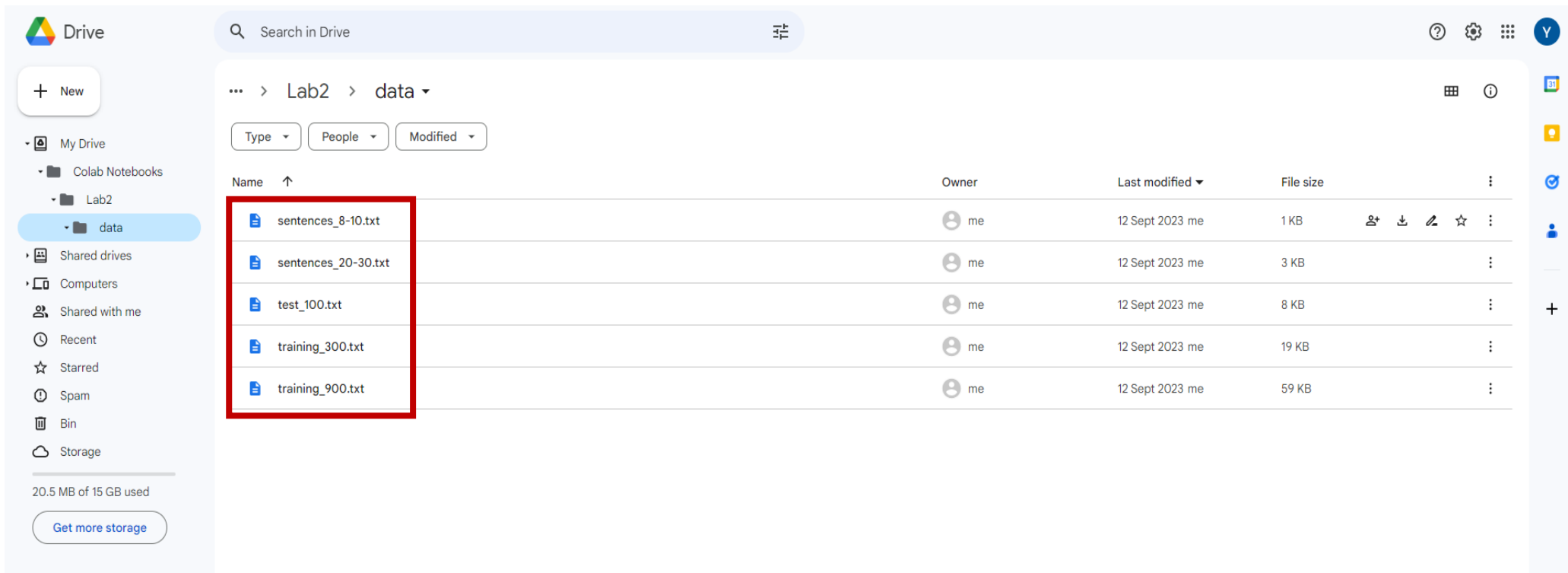
Prepare the workplace on Colab

- Open the “Colab Notebooks” folder and open the “Lab2” folder inside it. You should see a folder named “data” and a file named “NMT_man2can.ipynb”.



Prepare the workplace on Colab

- Inside the “data” folder, there should be five files, namely “sentences_8-10.txt”, “sentences_20-30.txt”, “test_100.txt”, “training_300.txt”, and “training_900.txt”.

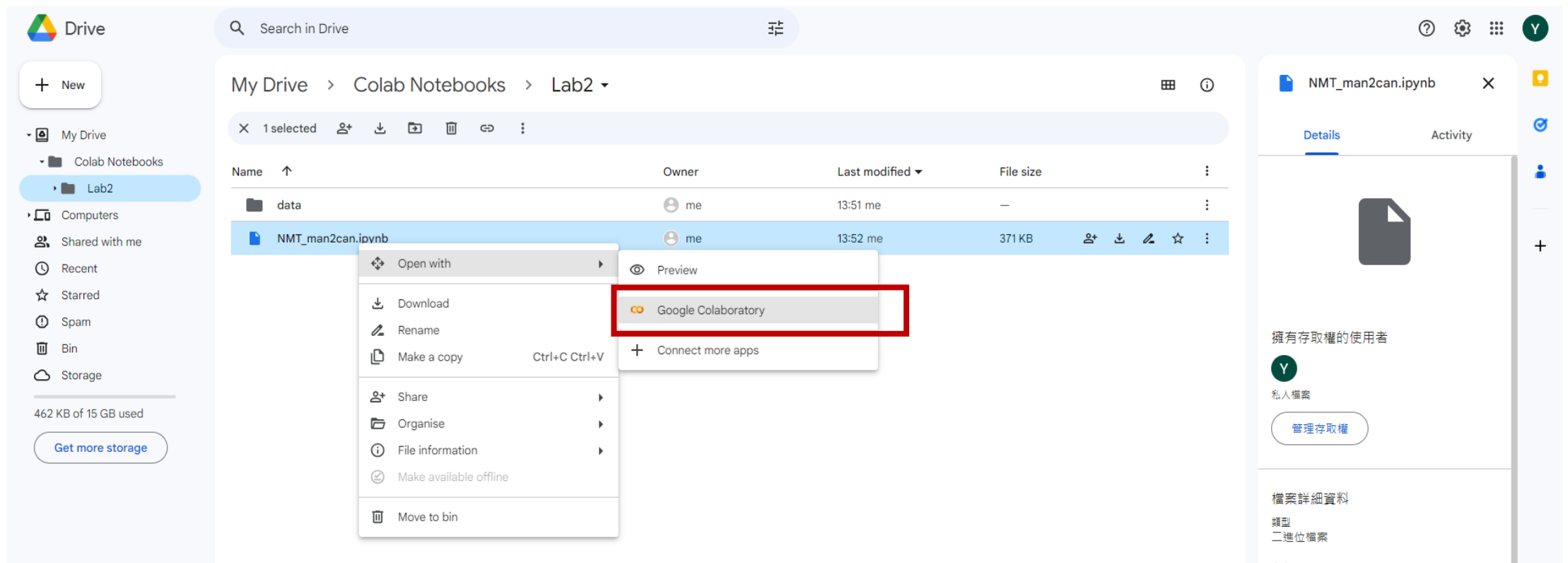


The screenshot displays the Google Drive web interface. On the left sidebar, the 'data' folder is selected under the 'Lab2' directory. The main panel shows a list of five text files within the 'data' folder, which are highlighted by a red rectangular box. The files are: 'sentences_8-10.txt' (1 KB), 'sentences_20-30.txt' (3 KB), 'test_100.txt' (8 KB), 'training_300.txt' (19 KB), and 'training_900.txt' (59 KB). All files are owned by 'me' and were last modified on '12 Sept 2023'.

Name	Owner	Last modified	File size
sentences_8-10.txt	me	12 Sept 2023	1 KB
sentences_20-30.txt	me	12 Sept 2023	3 KB
test_100.txt	me	12 Sept 2023	8 KB
training_300.txt	me	12 Sept 2023	19 KB
training_900.txt	me	12 Sept 2023	59 KB

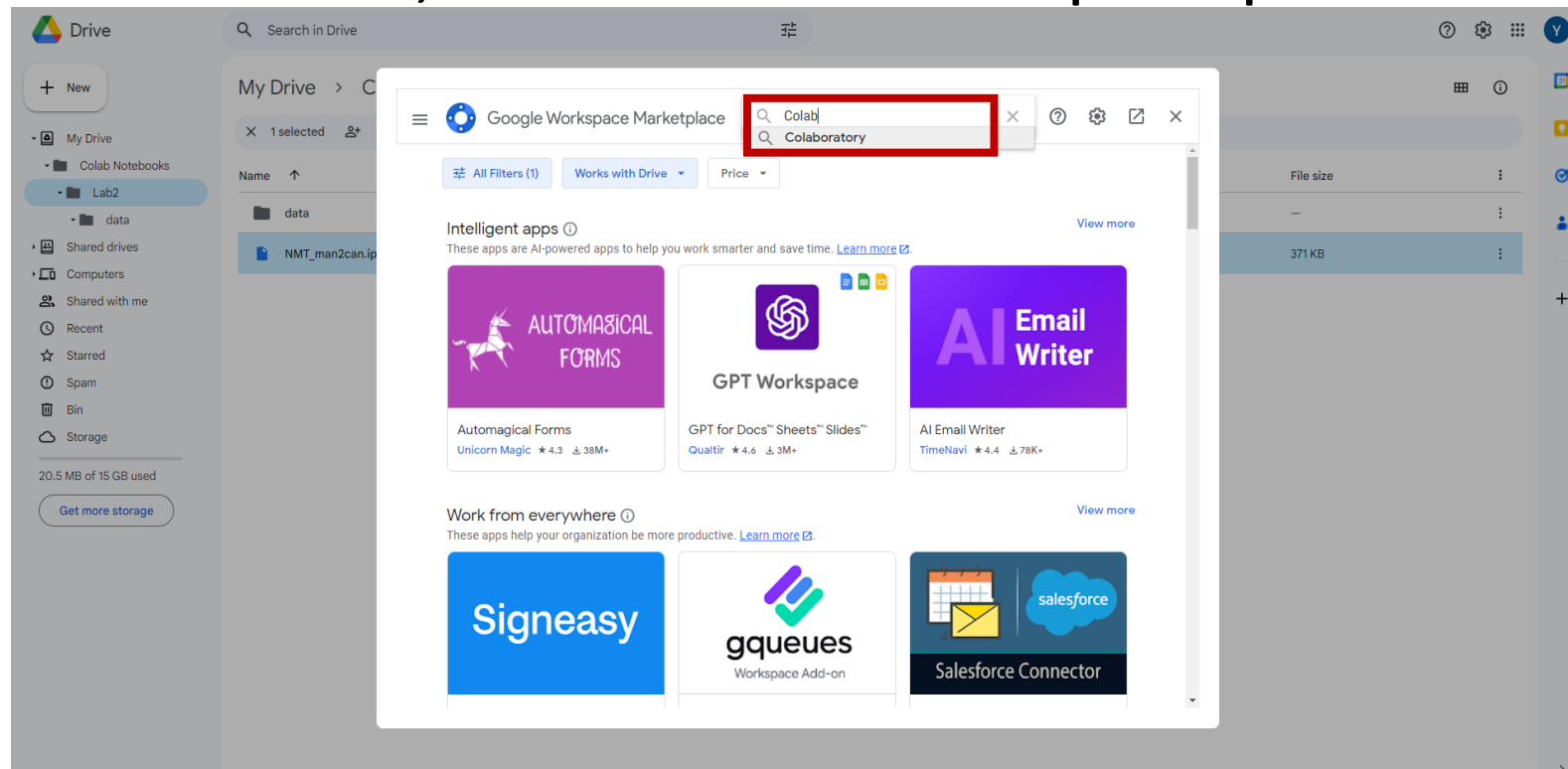
Prepare the workplace on Colab

- Right-click on the “NMT_man2can.ipynb” file and left-click to select “Connect more apps”.

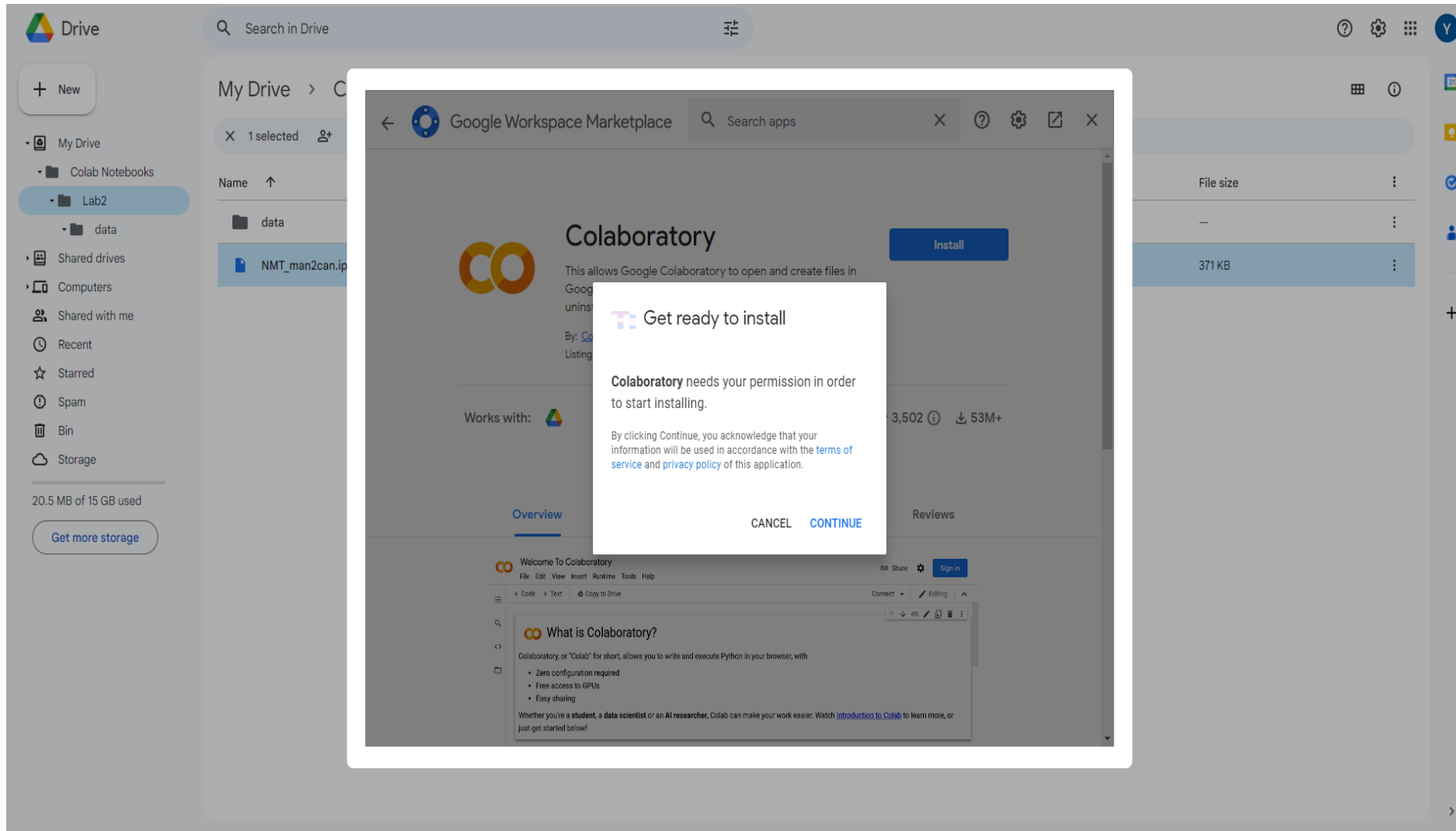


Prepare the workplace on Colab

- If you cannot see the option “google Colaboratory”, please click “Connect more apps”, type “Colaboratory” in the search box to search for the tool, or click on it in the prompt below the search box.

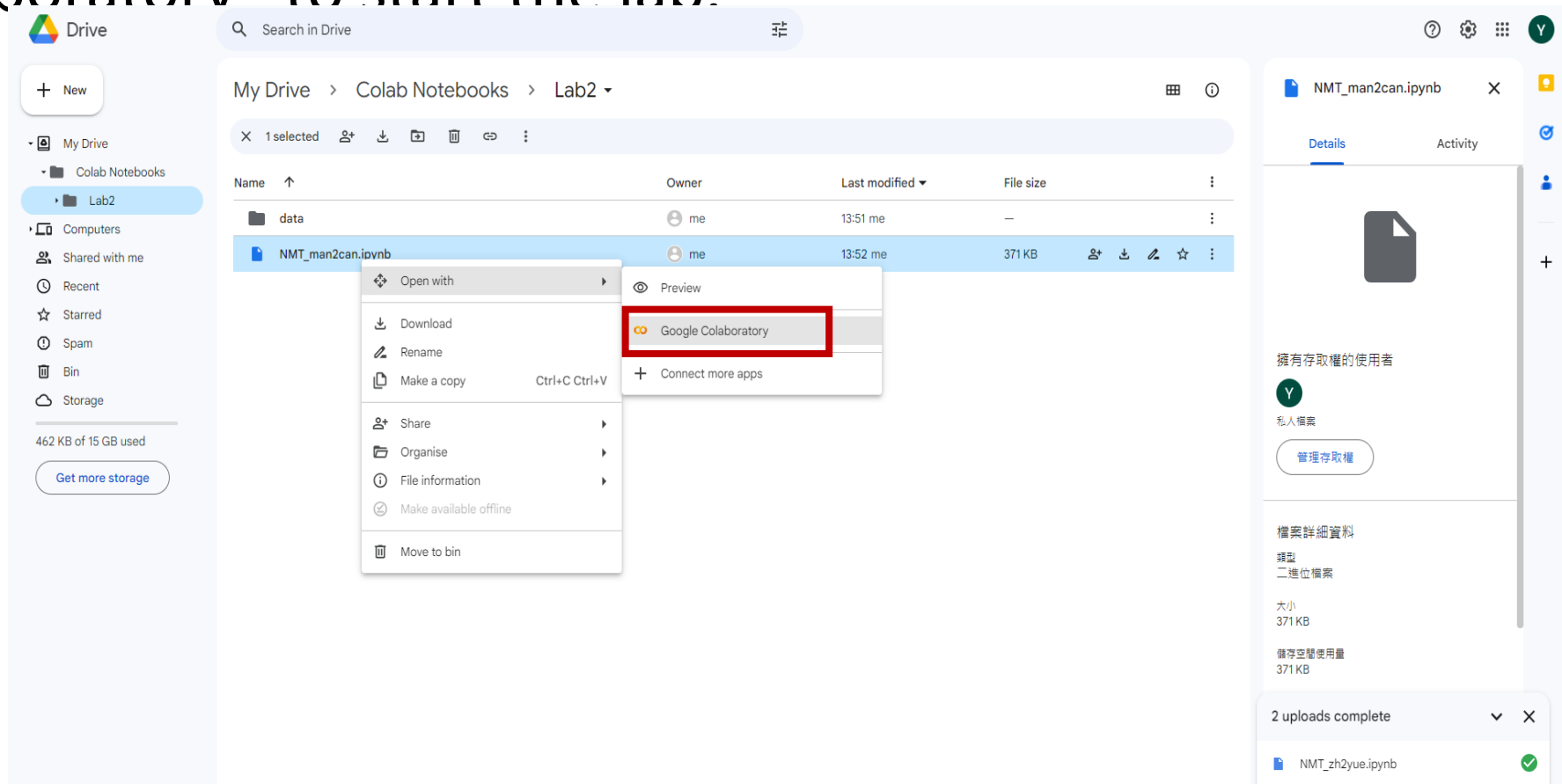


Install Google Colab on your browser



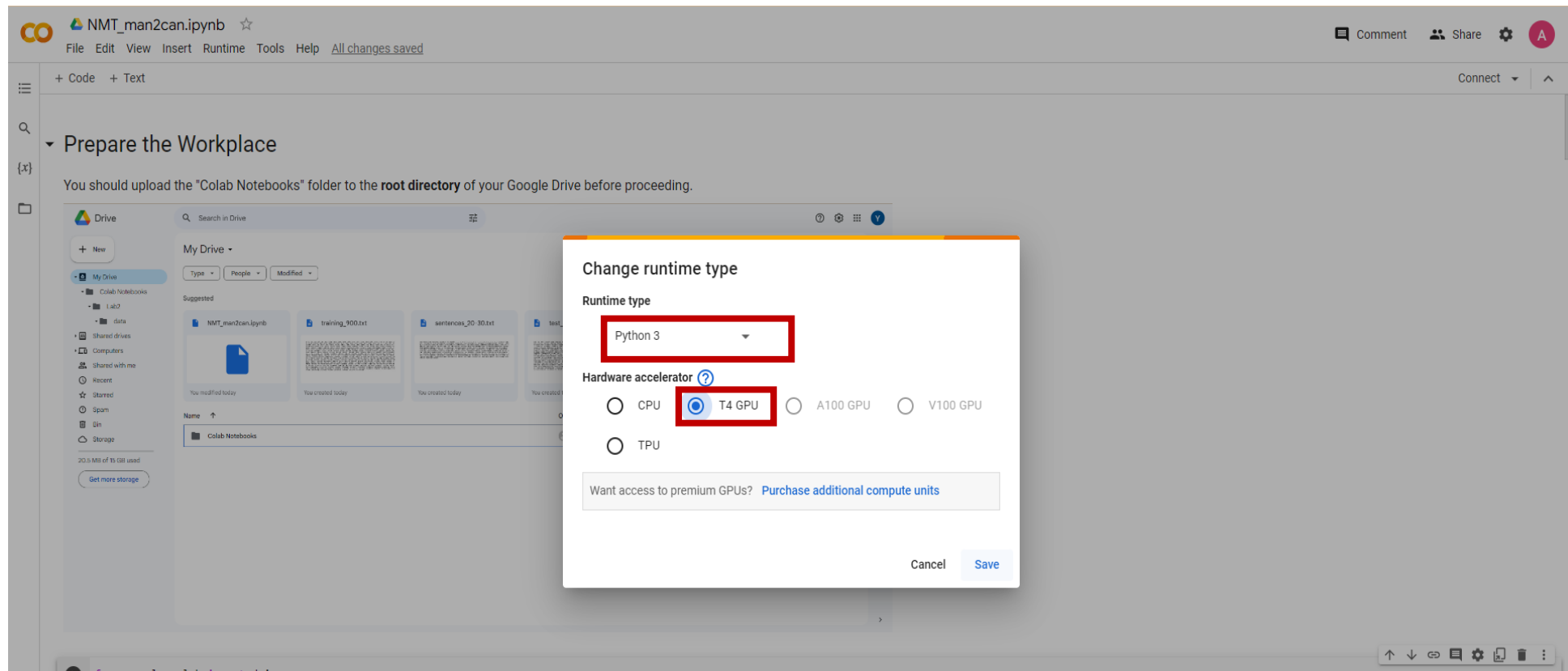
Prepare the workplace on Colab

- Right-click the “NMT_man2can.ipynb” file again and select “Google Colaboratory” to start the lab.



Setup of Colab

- In the Colab interface, click “Runtime” at the top-left menu bar and select “Change runtime type” in the prompted menu.



Initialize Drive

- New pop-up windows will be prompted to request your permission to access your Google Drive. Allow it to access and go through the login process again if required.

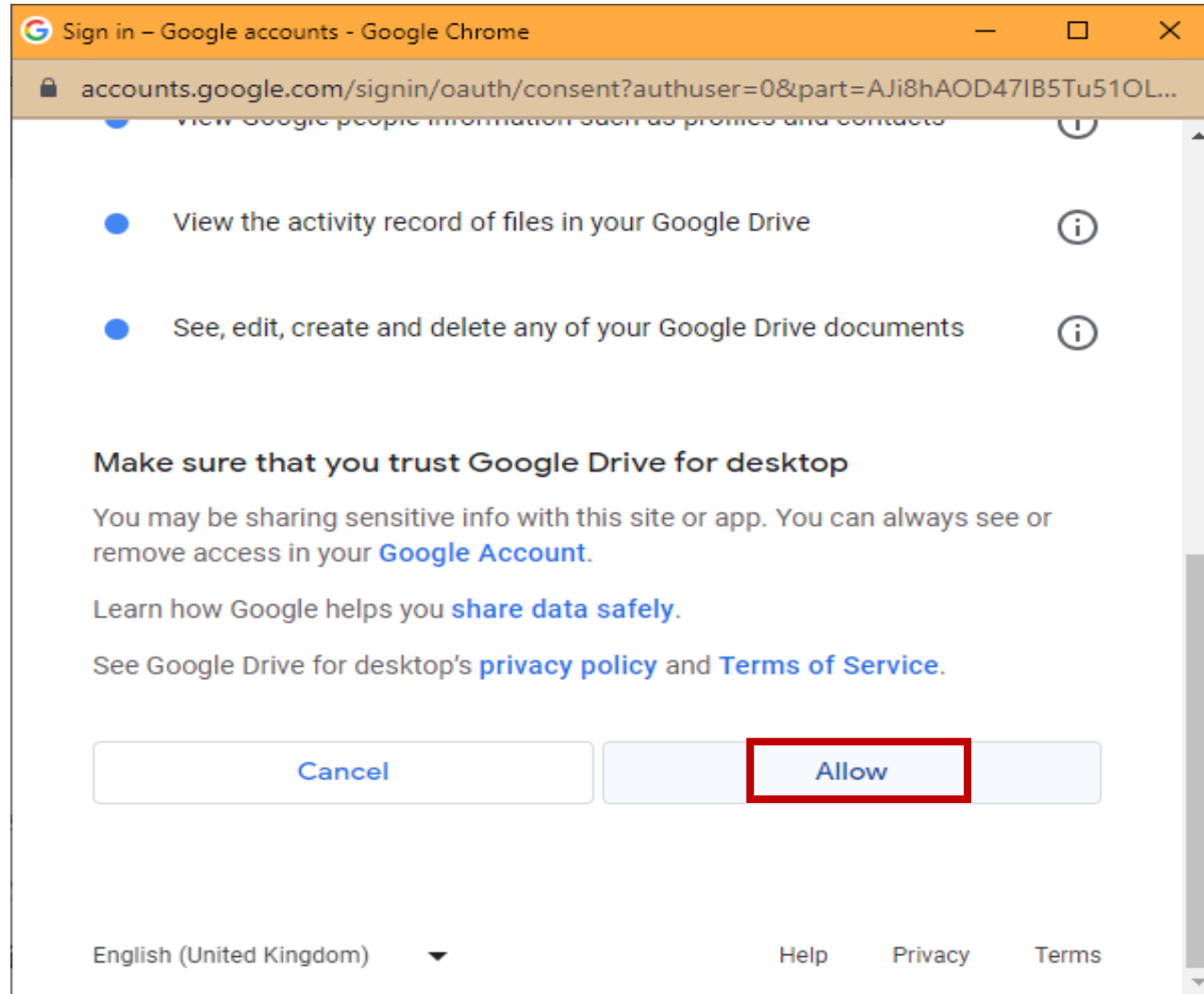
 31s 

```
from google.colab import drive
drive.mount('/content/drive')
%cd /content/drive/My Drive/Colab Notebooks/Lab2
```

 Mounted at /content/drive
/content/drive/My Drive/Colab Notebooks/Lab2

- ▼ Install Necessary Libraries and Import a Pre-trained Model

Initialize Drive



A toy NMT: man2can

- Standard Written Mandarin (man) vs. Cantonese (can)
- Example:
 - Is it theirs?: 是不是他們的？ vs. 係唔係佢哋嘅？
 - 茶餐廳 (Tea Restaurant)
 - zh.wikipedia: 茶餐廳是一種起源於香港的食肆，提供港式西餐及部分中式食品，是香港及澳門平民化的飲食場所。
 - zh-yue.wikipedia: 茶餐廳係一種香港起源嘅快餐食肆，提供揉合咗香港特色嘅西式餐飲同埋部分粵式食品，係香港平民化嘅飲食場所。
- Our demo:
 - Convert Standard Written Mandarin to Written Cantonese: **man2can**

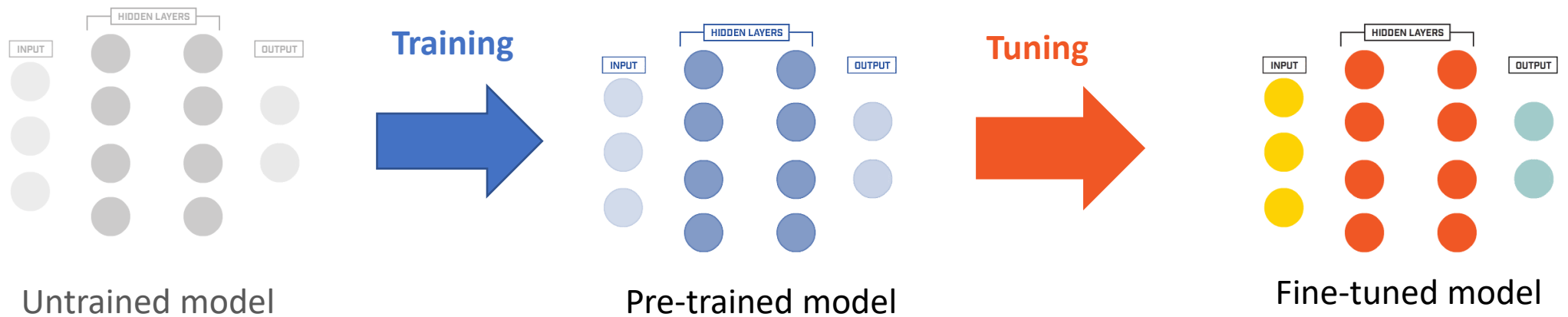


Neural machine translation (NMT)

- Machine translation: Automatically convert source text in one language to text in another language
- Neural machine translation: Use neural network model to learn the text conversion



Pre-trained and fine-tuned Model



Pretrained model and fine-tuning

- A pretrained model is a saved network that was previously trained on a **large dataset**. The English-language BERT was pretrained on BooksCorpus with 800M words and English Wikipedia with 2,500M words
- Fine-tuning is a process that takes a pretrained model and then tunes the model with a custom dataset

Deploy Pretrained Model

- Run the cells one by one
- After running the last cell, the following content will print out

Downloading: 100%  110k/110k [00:00<00:00, 1.13MB/s]

Downloading: 100%  29.0/29.0 [00:00<00:00, 653B/s]

Downloading: 100%  269k/269k [00:00<00:00, 1.34MB/s]

Downloading: 100%  624/624 [00:00<00:00, 15.7kB/s]

Downloading: 100%  412M/412M [00:12<00:00, 30.2MB/s]

```
Some weights of the model checkpoint at bert-base-chinese were not used when initializing BertModel:
- This IS expected if you are initializing BertModel from the checkpoint of a model trained on another task
- This IS NOT expected if you are initializing BertModel from the checkpoint of a model that you expect to be initialized from this checkpoint
Some weights of the model checkpoint at bert-base-chinese were not used when initializing BertLMHeadModel:
- This IS expected if you are initializing BertLMHeadModel from the checkpoint of a model trained on another task
- This IS NOT expected if you are initializing BertLMHeadModel from the checkpoint of a model that you expect to be initialized from this checkpoint
Some weights of BertLMHeadModel were not initialized from the model checkpoint at bert-base-chinese
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and other applications
```


Load Dataset

- Run the cells one by one
- man vs. can:

```
x_train[:10]
```

```
['只有她沒有回答問題。',  
'我無法表達我對你的感謝。',  
'我哭了一整晚。',  
'我想學跳舞。',  
'你別……別這樣和我說話。',  
'每一個字的意思都是獨特的。',  
'所以我叫你不要一個人去咯。',  
'你的手錶準不準?',  
'你醉了!',  
'我一定要把香菸戒了!']
```

```
y_train[:10]
```

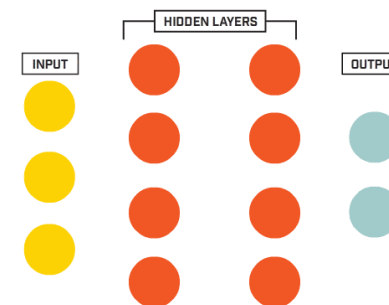
```
['淨係得佢無答到條問題。',  
'真係唔知點多謝你好。',  
'我喊咗成晚。',  
'我想學跳舞。',  
'你唔好……唔好噉同我講嘢。',  
'每一個字嘅意思都係獨特嘅。',  
'所以我咪叫你唔好一個人去囉。',  
'你隻錶準唔準㗎?',  
'你醉喇!',  
'我一定會戒煙!']
```

Token

Natural language
“插水”



Tokens
[102, 302, 493, 115]



Output
“突然变差”

Tokenization

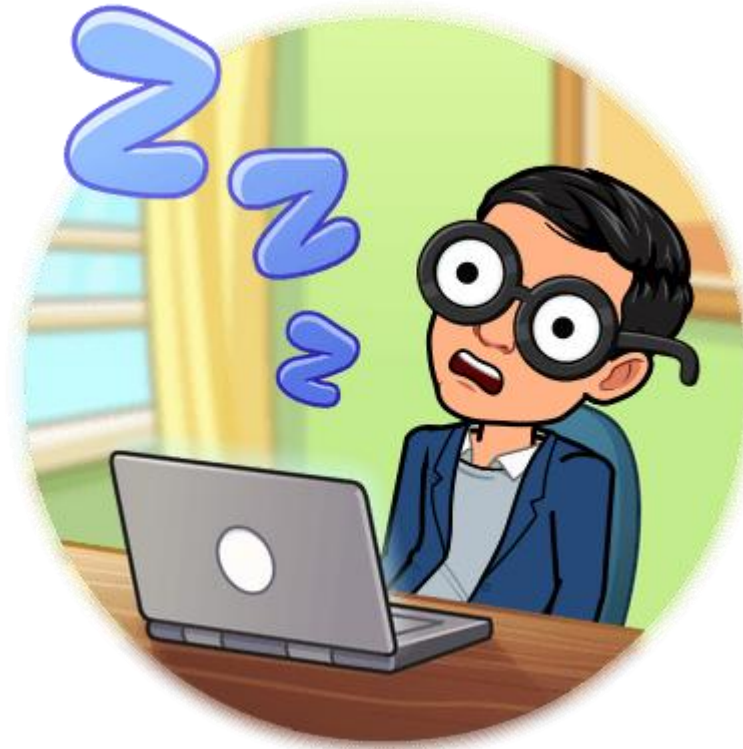
- Convert the text into numbers or the format which the model can understand
- Command in the demo:
 - `encoded = tokenizer.encode_plus(x, max_length = MAX_LENGTH, padding='max_length', truncation=True, return_tensors='pt')`

Example:

- “我認為大家都應該學另外一種語言。” (“I think we should all learn another language.”)
- Encoded = [101, 9970, 20173, 13671, 7723, 8362, 21784, 9899, 20146, 8302, 6393, 7706, 4896, 16056, 20190, 20037, 3566, 102, 0, 0, 0, 0, 0, 0]

Task 1

- **Task 1**: Obtain the encoded tokens of “然後我很快就睡著了。” (“After that, I soon fell asleep.”)



Fine-tuning

- Run the cells one by one
- The training loss should decrease with more training epochs
- After fine-tuning is done, the final model will be saved under the directory “/Colab Notebooks/Lab2/final_model_300”. The “**300**” here indicates that 300 pairs of Mandarin and Cantonese sentences are used to fine-tune the model.

```
-----
Epoch 18/20
100% ██████████ 19/19 [00:03<00:00, 5.75it/s]
Loss: 0.21422484673951803
-----
Epoch 19/20
100% ██████████ 19/19 [00:03<00:00, 5.76it/s]
Loss: 0.2026648509659265
-----
Epoch 20/20
100% ██████████ 19/19 [00:03<00:00, 5.74it/s]
Loss: 0.2084158109991174
-----
Model saved at folder /content/drive/MyDrive/Colab Notebooks/Lab2/final_model_300.
-----
```

✓ 1m 30s completed at 16:08

Evaluation

- A test sentence: “她是一個非常美的女人。”
- Compare the output of the following models:
 - *pretrained_model*
 - *final_model_300*
- Compute the **BLEU** (Bi-Lingual Evaluation Understud score for the test sentence with the reference, “她是一個非常美的女人。”
 - A BLEU score is a number between 0 and 1, with closer to 1 representing the better translation

After fine-tuning:

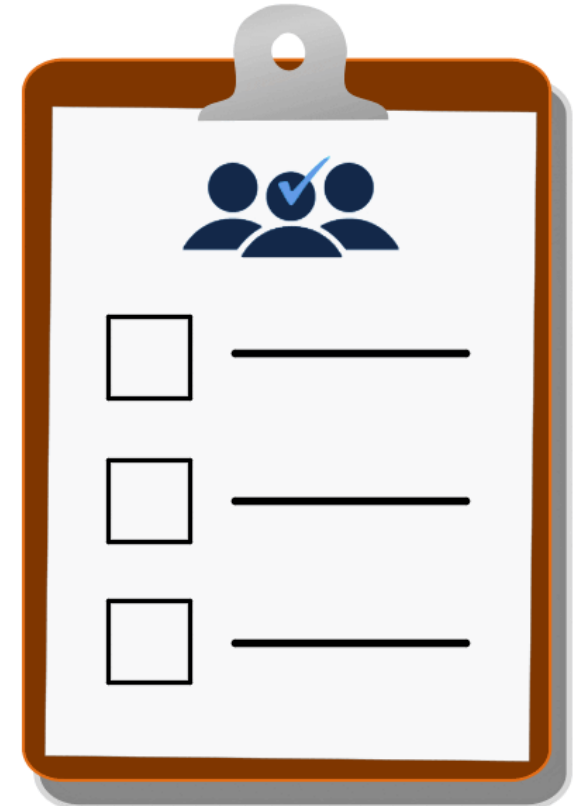
```
✓ [15] translation = translate_sentence(test_sentence, final_model,
                                     tokenizer, device, MAX_LENGTH)
      translation = ' '.join(list(translation)).split()
      print("Translation:", ' '.join(translation))

      score = sentence_bleu([list(reference)], translation,
                           smoothing_function=SmoothingFunction().method4)
      print("BLEU score:", score)
```

Again, you can select and copy from the above sentences to [Bing Microsoft Tran](#) English for reference only if you wish.

BLEU

- **BLEU**: Bi-Lingual Evaluation Understudy score
- A number between 0 and 1, with closer to 1 representing the better translation
- Task 1: Test the sentence “我哋可以食飯了么？”



Task 2: Evaluate the performance of the model before and after fine-tuning

- Translate the Mandarin sentence “她是一個非常美的女人。” using the two models and report the BLEU scores of both translations.

After fine-tuning:

```
✓ [15] translation = translate_sentence(test_sentence, final_model,
                                     tokenizer, device, MAX_LENGTH)
    translation = ' '.join(list(translation)).split()
    print("Translation:", ' '.join(translation))

    score = sentence_bleu([list(reference)], translation,
                          smoothing_function=SmoothingFunction().method4)
    print("BLEU score:", score)
```

Again, you can select and copy from the above sentences to [Bing Microsoft Tran](#)
English for reference only if you wish.

- Use the same test set, “**test_100.txt**”, to evaluate and compare the pre-trained model and “**final_model_300**” by checking the average BLEU scores for each case. **Report the two BLEU scores for the pre-trained model and the model “final_model_300”.**

Task 3: Evaluate the performance of the fine-tuned “final_model_300”

- **Sentence Length (SL)**: number of characters of the sentence. E.g., the length of “然後我很快就睡著了。” (“After that, I soon fell asleep.”) is 10: “然”, “後”, “我”, “很”, “快”, “就”, “睡”, “著”, “了”, “。”.
- Refer to another two test sets, each containing 20 sentences.
 - sentences_8-10.txt: SL ranges from 8 to 10.
 - sentences_20-30.txt: SL ranges from 20 to 30
- **Report the two average BLEU scores on the two test sets and see if SL affects the model’s performance.**

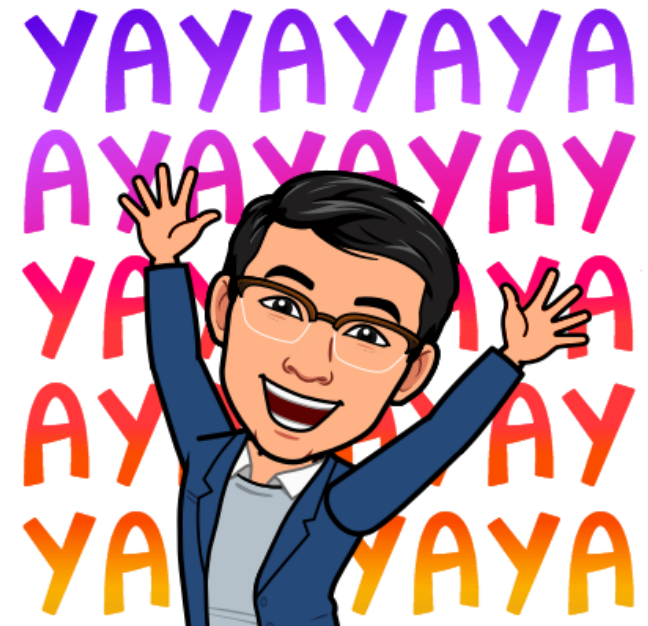
Task 4: Evaluate the performance of models trained with different amounts of data.

- Train another model with 900 sentences, “**final_model_900**”
- Obtain the output of the newly fine-tuned model using the same example test sentences and report the BLEU score.
- Use the same test set (**test_100.txt**) to evaluate the model and report the average BLEU score.
- Compare the BLEU scores with the ones that you obtained in Task 2 using “**final_model_300**”. Observe if the two models have different performance. If yes, try to explain why.

Task 5: Train a NMT model for can2man.

- The NMT model in the demo is used to convert Mandarin to Cantonese.
- With the same model structure and parameters, train a model used to **convert Cantonese to Mandarin** and **report the performance of the can2man model**.

Task 5 is optional



Task 5 (Optional)

- Train an NMT model for *yue2zh*
 - The NMT model in the demo is used to convert *zh* to *yue*. With the same model structure and parameters, train a model used to convert *yue* to *zh*, and report the performance of *yue2zh* model